Channabasava Chola

Researcher in Computer Vision and Machine Learning
PhD Research Scholar, Department of Studies in Computer Science, University of Mysore, Mysuru, Karnataka, India
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Research Profile

Computer vision and machine learning researcher with six years of research experience, working on medical and microscopic imaging, IoMT-based diagnostics and reinforcement learning for robotic grasping. Author of 21 publications (5 as first author) and two Best Paper awards.

Education

Doctor of Philosophy in Computer Science *Aug 2024 – Present*
Department of Studies in Computer Science, University of Mysore, Mysuru, India

- Research direction: IoMT-based diagnostic and point-of-care systems.

Master of Computer Applications *Jul 2017 – Aug 2019*
University of Mysore, Mysuru, India

- Research direction: machine learning and image processing.
- Thesis: "Drosophila melanogaster Gender Classification by Microscopic Images".

Bachelor of Science (Computer Science, Mathematics, Physics) *Jul 2014 – Jun 2017*
Gulbarga University, Gulbarga, India

Research Experience

Research Scholar *Aug 2024 – Present*
Department of Studies in Computer Science, University of Mysore, Mysuru, India

- Developing computer-aided diagnostic (CAD) systems.
- Developing reinforcement learning approaches for robotic grasping in the MuJoCo simulator.

Research Assistant *Sep 2022 – Aug 2023*
Kyung Hee University, Yongin-si, Gyeonggi-do, Republic of Korea

- Developed 3D object detection with hand–eye calibration for object grasping with an Omron TM robot arm.
- Developed reinforcement learning approaches for robotic grasping in the MuJoCo simulator.
- Developed human activity forecasting models.

Project Associate *Mar 2021 – Aug 2022*
CSIR–Central Food Technological Research Institute (CSIR–CFTRI), Mysuru, India

- Developed 2D image classification and ripeness detection for mango images.
- Developed BCNet, a deep learning framework for human blood cell classification.
- Developed FMDNet, a deep learning model for face mask detection to help limit COVID-19 infection.

Teaching Assistant *Sep 2019 – Mar 2021*
Indian Institute of Information Technology Kottayam (IIIT-K), Kerala, India

- Developed deep learning models for sunspot detection and classification.
- Developed a deep learning system for COVID-19 detection.
- Worked on Arabic text classification with machine learning.
- Worked on microscopic image classification for Drosophila gender identification.

Teaching and Mentoring

Visiting Professor *Sep 2025 – Oct 2025*
Xuchang University, Xuchang, Henan Province, China

- Delivered lectures on Software Engineering and C++.

Visiting Professor *Apr 2025 – Jun 2025*
Huanghuai University, Zhumadian, Henan Province, China

- Delivered lectures on Advanced Java, C++ and Human–Computer Interface.

Assistant Professor *Oct 2024 – Mar 2025*
Department of AI & ML, Mysore University School of Engineering, University of Mysore, India

- Taught Deep Learning and Cybersecurity to B.Tech AI & ML and AI & DS students.
- Guided undergraduate student projects.

Visiting Professor *Oct 2024 – Dec 2024*
Huanghuai University, Zhumadian, Henan Province, China

- Delivered lectures on Advanced Java, C++ and Artificial Intelligence.

Visiting Professor *Jun 2024 – Jul 2024*
Huanghuai University, Zhumadian, Henan Province, China

- Delivered lectures on Advanced Java and C++.

Guest Lecturer *Dec 2023 – Present*
Dr. Gangubai Hangal Music and Performing Arts University, Mysuru, India

- Teaching Digital Fluency and Cybersecurity to undergraduate students.

Thesis Supervisor *Sep 2022 – Present*
upGrad, India

- Guiding cohorts of data science and machine learning students through their theses.

Publications

21 publications; 5 as first author. Google Scholar: profile.

Journal Articles

[1] M. Masadeh, **C. Chola**, A. Y. Muaad. Investigating the Impact of Preprocessing Techniques and Representation Models on Arabic Text Classification using Machine Learning. *International Journal of Advanced Computer Science and Applications 15(1)*, 2024.

[2] I. E. Jaramillo, **C. Chola**, J.-G. Jeong, J.-H. Oh, H. Jung, J.-H. Lee, W. H. Lee, T.-S. Kim. Human Activity Prediction Based on Forecasted IMU Activity Signals by Sequence-to-Sequence Deep Neural Networks. *Sensors 23(14): 6491*, 2023. [link]

[3] J. V. Bibal Benifa, **C. Chola**, A. Y. Muaad, M. A. Bin Hayat, B. Bin Heyat, R. Mehrotra, F. Akhtar, H. S. Hussein, D. L. Ramírez Vargas, Á. Kuc Castilla, I. de la Torre Díez, S. Khan. FMDNet: An Efficient System for Face Mask Detection Based on Lightweight Model during COVID-19 Pandemic in Public Areas. *Sensors 23(13): 6090*, 2023.

[4] P. R. Lopez, J.-H. Oh, J. G. Jeong, H. Jung, J. H. Lee, I. E. Jaramillo, **C. Chola**, W. H. Lee, T.-S. Kim. Dexterous Object Manipulation with an Anthropomorphic Robot Hand via Natural Hand Pose Transformer and Deep Reinforcement Learning. *Applied Sciences 13(1): 379*, 2023. [link]

[5] **C. Chola**, A. Y. Muaad, M. B. Bin Heyat, J. V. Bibal Benifa, W. R. Naji, K. Hemachandran, N. F. Mahmoud, N. A. Samee, M. A. Al-Antari, Y. M. Kadah, T.-S. Kim. BCNet: A Deep Learning Computer-Aided Diagnosis Framework for Human Peripheral Blood Cell Identification. *Diagnostics 13(6): 1104*, 2023. [link]

[6] K. Hemachandran, A. Alasiry, M. Marzougui, S. M. Ganie, A. A. Pise, M. T.-H. Alouane, **C. Chola**. Performance Analysis of Deep Learning Algorithms in Diagnosis of Malaria Disease. *Diagnostics 13(3): 534*, 2023. [link]

[7] A. S. Alphonse, J. V. Bibal Benifa, A. Y. Muaad, **C. Chola**, M. B. Bin Heyat, B. A. H. Murshed, N. A. Samee, M. Alabdulhafith, M. A. Al-antari. A Hybrid Stacked Restricted Boltzmann Machine with Sobel Directional Patterns for Melanoma Prediction in Colored Skin Images. *Diagnostics 12(11): 2815*, 2022. [link]

[8] **C. Chola**, J. V. Bibal Benifa, D. S. Guru, A. Y. Muaad, J. Hanumanthappa, M. A. Al-antari, H. AlSalman, A. H. Gumaei. Gender Identification and Classification of Drosophila Melanogaster Flies using Machine Learning Techniques. *Computational and Mathematical Methods in Medicine (Hindawi)*, 2022. [link]

[9] L. M. Mestetskiy, J. V. Bibal Benifa, D. S. Guru, **C. Chola**, H. S. Nagendraswamy. Gender Identification of Drosophila Melanogaster based on Morphological Analysis of Microscopic Images. *The Visual Computer (Springer)*, 2022. [link]

[10] A. Y. Muaad, J. Hanumanthappa, D. S. Guru, J. V. Bibal Benifa, **C. Chola**, M. A. Al-antari, H. AlSalman, A. H. Gumaei. Performance Investigation of Pre-processing and Representation Techniques on Arabic Document Classification. *Mathematical Problems in Engineering (Hindawi)*, 2022. [link]

[11] L. M. Mestetskiy, D. S. Guru, H. S. Nagendraswamy, **C. Chola**. Recognition of Drosophila Sex from Photographic Images (English translation of the Russian title). *Ecology. Economics. Informatics. Series: Geoinformation Technologies and Space Monitoring, no. 4, pp. 80–84 (in Russian)*, 2019.

Conference Papers

[12] R. Nyange, M. G. Chipofya, S. Goel, **C. Chola**. Enhancing Maize Crop Health: Deep Learning Approach for Disease Detection and Classification Using Leaf Images. *4th International Conference on Artificial Intelligence and Knowledge Processing (AIKP 2024), Springer*, 2024.

[13] M. Pramodha, A. Y. Muaad, R. Nyange, A. Anusha, **C. Chola**. Explainable AI Based Framework for Banana Disease Detection. *5th International Conference on Innovative Trends in Information Technology (ICITIIT 2024), IEEE, pp. 1–6*, 2024.

[14] J. G. Jeong, J.-H. Oh, H. Jung, J. H. Lee, I. E. Jaramillo, **C. Chola**, W. H. Lee, T.-S. Kim. Long-Horizon Manipulation by a Single-arm Robot via Sub-goal Network based Hierarchical Reinforcement Learning. *ICBET 2023, Tokyo, Japan*, 2023. [link]

[15] **C. Chola**, J. V. Bibal Benifa. Detection and Classification of Sunspots via Deep Convolution Neural Networks. *International Conference on Intelligent Engineering Approach (ICIEA 2022)*, 2022. [link]

[16] A. Y. Muaad, J. Hanumanthappa, J. V. Bibal Benifa, **C. Chola**, M. Naveen Mourya, G. Hemantha Kumar, M. Pramodha. An Effective Approach for Arabic Document Classification Using Machine Learning. *International Conference on Intelligent Engineering Approach (ICIEA 2022)*, 2022. [link]

[17] **C. Chola**, M. Pramodha, A. Y. Muaad, J. V. Bibal Benifa, J. Hanumanthappa, M. Al-antari. A Hybrid Deep Learning Approach for COVID-19 Diagnosis via CT and X-ray Medical Images. *Computer Sciences & Mathematics Forum 2(1): 13, MDPI*, 2021. [link]

[18] **C. Chola**, M. B. Bin Heyat, F. Akhtar, O. Al Shorman, J. V. Bibal Benifa, A. Y. M. Muaad, M. Masadeh, F. Alkahatni. IoT Based Intelligent Computer-Aided Diagnosis and Decision Making System for Health Care. *IEEE ICIT 2021*, 2021. [link]

[19] A. Y. Muaad, M. A. Al-antari, **C. Chola**, J. V. Bibal Benifa, J. Hanumanthappa. AI-based Misogyny Detection from Arabic Levantine Twitter Tweets. *Proceedings of the 1st Online Conference on Algorithms (IOCA), MDPI, Basel, Switzerland*, 2021. [link]

[20] J. Hanumanthappa, A. Y. Muaad, J. V. Bibal Benifa, **C. Chola**, V. Hiremath, M. Pramodha. IoT-Based Smart Diagnosis System for Health Care. *Sustainable Communication Networks and Application, Springer, Singapore, pp. 1–15*, 2021. [link]

Book Chapters

[21] J. V. Bibal Benifa, J. Philip, **C. Chola**. Detection and Classification of Brain Tumour from MRI Images by Different Classifiers. *Chapter 7 in High-Performance Medical Image Processing, Apple Academic Press (ISBN 9781774637227)*.

Under Review

- R. Pal, F. Akhtar, M. B. Bin Heyat, **C. Chola**, P. K. Patel, O. Al Shorman, A. Alexiou, Z. You. A Comprehensive Survey Based Study Utilizing Systematic Literature Review and Network Visualization Techniques with Special Emphasis on Effect, Intervention, Treatment Planning, and Impact of Yoga in Stress: Current Advancements and Future Directions. *BioMed Research International (Hindawi), under review*.

Awards

- **Feb 2022.** Best Paper Award, International Conference on Intelligent Engineering Approach (ICIEA 2022).
- **2021.** Best Paper Award, 1st Online Conference on Algorithms (IOCA 2021), Artificial Intelligence Algorithms session, MDPI.

Skills

Research areas	Machine learning, deep learning, image processing, natural language processing, reinforcement learning
Programming	Python, MATLAB
Simulation and hardware	MuJoCo simulator, Omron TM robot arm (hand–eye calibration)
Tools	LaTeX, Mendeley, Microsoft Office
Languages	English, Hindi, Kannada, Telugu, Marathi

Conference Activity and Service

- Presented a paper at ICBET 2023, Tokyo, Japan.
- Participated in KICS 2022, Gyeongju-si, Republic of Korea.
- Session chair, 10th International Conference on Information Technology (ICIT 2021), Day 1, 14 July 2021.
- Poster presentation: “Deep Learning for Heliophysics”, IIAP 2021, Bangalore (Advances in observations and modelling of solar magnetism and variability).
- Volunteer, International Conference on Data Analytics and Learning (DAL 2018), University of Mysore.
- Volunteer, ICITIIT 2020 and ICITIIT 2021, IIIT Kottayam.
- National Service Scheme (NSS) volunteer, Government First Grade College, Bidar.
- National Cadet Corps (NCC) “C” certificate holder, 3 KAR AIR (T) SQN, Bidar.

References

Available on request.